

Bayes Decision Theory

Notes for Probabilistic Models of the Visual Cortex

Alan Yuille

Johns Hopkins University

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What this lecture covers

Bayes decision theory

Probabilities

The Bayes rule

Signal detection theory

Learning the distributions

The limits of BDT

Summary

The three ingredients

Bayes decision theory (BDT) is built from three objects.

- ▶ A joint distribution $P(x, y)$ over observations x and the quantity y we want to know.
- ▶ A set $\mathcal{A}$ of decision rules α , each mapping an observation to a decision $\alpha(x)$.
- ▶ A loss $L(\alpha(x), y)$, the cost of deciding $\alpha(x)$ when the truth is y .

Only the first is empirical. The other two are choices you must make.

In vision, x is what the sensor delivers, an image or a filter response, and y is a property of the scene: the object present, the label of a pixel, the depth at a point.

The three ingredients

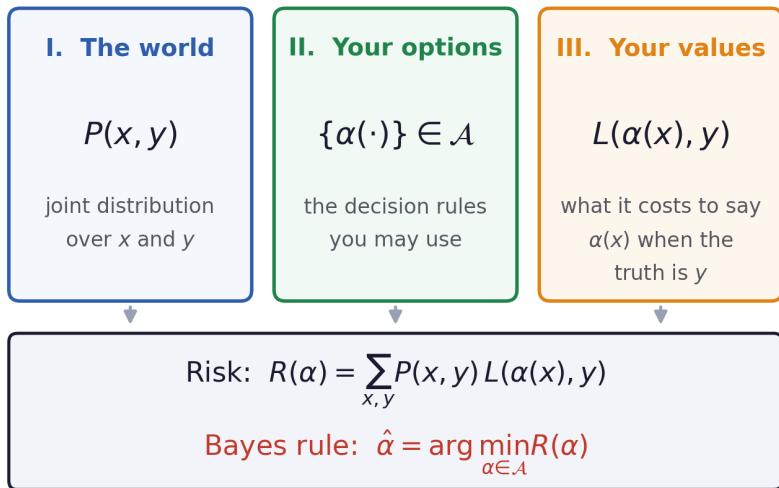


Figure 1: The three ingredients of Bayes decision theory. The joint distribution is a fact about the world; the rule class and the loss are chosen by the person posing the problem.

Risk and the Bayes rule that minimises it

The risk of a rule is its loss averaged over the joint distribution,

$$R(\alpha) = \sum_x \sum_y L(\alpha(x), y) P(x, y),$$

and for continuous observations the sum over x becomes an integral,

$$R(\alpha) = \int \sum_y L(\alpha(x), y) P(x, y) dx.$$

The Bayes rule is the minimiser over the admissible rules,

$$\hat{\alpha} = \arg \min_{\alpha \in \mathcal{A}} R(\alpha), \quad \text{and } R(\hat{\alpha}) \text{ is the Bayes risk.}$$

The risk is the average cost of a rule over all examples. The Bayes risk is never zero, because decisions cannot be perfect: whenever two classes can produce the same observation, some observations are decided wrongly whatever rule is used. Every visual task can be formalized with this formula, evaluated with different losses.

An example of risk and decision rules

Take three possible observations x_1, x_2, x_3 and two classes, with the joint probabilities

	x_1	x_2	x_3	
$y = +1$	0.20	0.08	0.02	$P(y = +1) = 0.30$
$y = -1$	0.05	0.15	0.50	$P(y = -1) = 0.70$

A false positive costs 1 and a false negative costs 4. Correct decisions cost nothing, so only one cell of each column contributes.

For the rule $\alpha = (+1, -1, -1)$ the three contributions are $0.05 \times 1 = 0.05$, then $0.08 \times 4 = 0.32$, then $0.02 \times 4 = 0.08$, giving $R(\alpha) = 0.45$.

An example of risk and decision rules

	x_1	x_2	x_3
$y = +1$	0.20	0.08 loss 4	0.02 loss 4
$y = -1$	0.05 loss 1	0.15	0.50
cost	0.05	0.32	0.08

$$R(\alpha) = 0.45$$

Figure 2: One rule and its risk. Each column contributes the probability of the class the rule gets wrong, times the cost of that mistake.

The risk of each of the eight rules

With three observations and two decisions there are $2^3 = 8$ rules. Deciding $+1$ at x_j costs $P(x_j, y = -1) \times 1$ and deciding -1 costs $P(x_j, y = +1) \times 4$:

	x_1	x_2	x_3
cost of deciding $+1$	0.05	0.15	0.50
cost of deciding -1	0.80	0.32	0.08

The columns are independent of one another, so taking the cheaper entry in each column gives $\hat{\alpha} = (+1, +1, -1)$ with Bayes risk $0.05 + 0.15 + 0.08 = 0.28$.

The risk of each of the eight rules

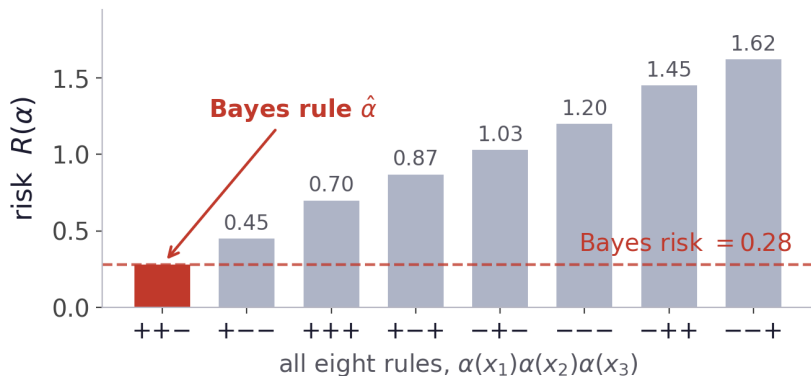


Figure 3: The Bayes rule is the one with the smallest risk among the 2^3 decision rules. Imperfection of decisions keeps that risk above zero.

Two ways to represent the joint probability

The joint distribution factorises in two ways,

$$P(x, y) = P(x | y) P(y) = P(y | x) P(x),$$

and equating them gives Bayes' theorem,

$$P(y | x) = \frac{P(x | y) P(y)}{P(x)}.$$

The denominator $P(x)$ just normalises it, it does not affect which y is probable.

Two ways to represent the joint probability

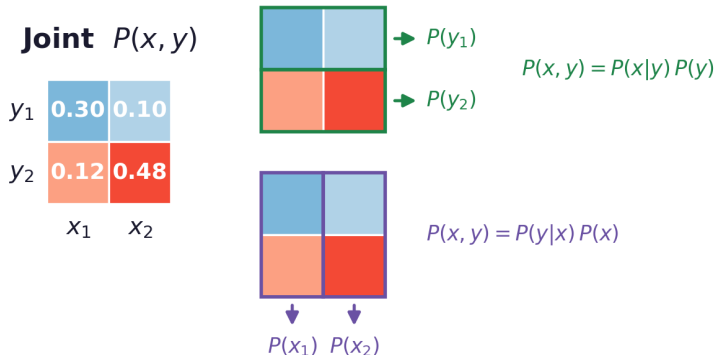


Figure 4: A joint distribution read two ways. Going down the columns gives the likelihood and the prior; going along the rows gives the posterior and the evidence.

Likelihood, prior, posterior

- ▶ $P(x|y)$ is what this observation says about y .
- ▶ $P(y)$ is what you knew independently of the observation.
- ▶ $P(y|x)$ is your belief after combining the two.

The likelihood is a function of y for the x you saw, so it need not sum to one over y . The prior and the posterior are distributions over y and do.

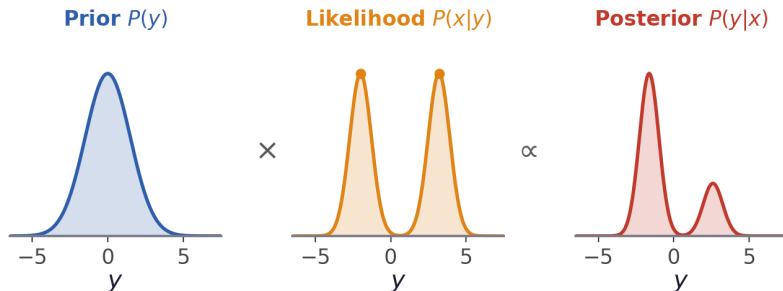


Figure 5: Two peaks in the likelihood are equally good explanations of the observation. The prior decides between them, and the posterior keeps the one the prior supports.

The evidence term is a sum over the classes

The denominator is fixed by the numerator, because the posterior sums to one over y :

$$P(x) = \sum_y P(x | y) P(y) \implies \sum_y P(y | x) = 1.$$

For two classes this is

$P(x) = P(x | y = +1)P(y = +1) + P(x | y = -1)P(y = -1)$, and the posterior for the target is

$$P(y = +1 | x) = \frac{P(x | y = +1) P(y = +1)}{P(x | y = +1) P(y = +1) + P(x | y = -1) P(y = -1)}.$$

$P(x)$ is the same number for every class, so it cancels from any comparison between two classes. It matters when the absolute value of the posterior is wanted, and in model comparison, where $P(x)$ under two different models is the quantity being compared.

Why inverse problems need prior

In vision, $P(I | W)$ is the probability of generating an image I from a world state W . The forward process is what you can specify by computer graphics: geometry, lighting, projection and noise. The inverse $P(W | I)$ is the posterior distribution of the world state, and it is ill-posed:

$$P(W | I) \propto P(I | W) P(W).$$

Many world states render to the same image. A grey patch is a white surface in shadow or a grey surface in light; a small image region is a near small object or a far large one. The likelihood gives all of them the same value.

The prior $P(W)$ supplies what the image alone leaves undetermined, making the inverse problem well posed. Surfaces are usually smooth, illumination usually comes from above, objects usually rest on supporting surfaces.

The rule depends only on the posterior

Rewrite the risk with $P(x, y) = P(y | x)P(x)$ and group the terms by observation:

$$R(\alpha) = \sum_x P(x) \underbrace{\left[\sum_y L(\alpha(x), y) P(y | x) \right]}_{\text{the cost of the decision made at } x} .$$

The inner sum minimises separately for each x , so the decision only depends on x . Specifically, it only depends on the posterior distribution $P(y | x)$.

Many researchers usually learn $P(y | x)$ directly and skip the prior $P(y)$ and likelihood $P(x | y)$. The grouping above licenses that: the posterior is a sufficient input to the decision.

Minimising the risk one observation at a time

Write $C(a|x) = \sum_y L(a, y) P(y|x)$ for the posterior expected cost of answering a at the observation x . The risk is then

$$R(\alpha) = \sum_x P(x) C(\alpha(x)|x) \geq \sum_x P(x) \min_a C(a|x),$$

because $P(x) \geq 0$ and each term is bounded below by the smallest value its bracket can take. The bound is attained by the rule that picks the minimiser at every observation,

$$\hat{\alpha}(x) = \arg \min_a \sum_y L(a, y) P(y|x).$$

Two consequences follow. The Bayes rule is deterministic: randomising over answers can only average values that are all at least the minimum. And the rule is local: what it answers at x is unaffected by the other observations, as long as $\mathcal{A}$ contains every mapping.

The two special cases: MAP and ML

With the 0-1 loss, $L(a, y) = 1 - \delta_{a,y}$, the posterior expected cost is $C(a | x) = 1 - P(y=a | x)$, and minimising it maximises the posterior:

$$\hat{\alpha}(x) = \arg \max_y P(y | x) \quad \text{MAP (maximum a posteriori).}$$

If in addition the prior is uniform, $P(y)$ is a constant that leaves the ranking unchanged, and

$$\hat{\alpha}(x) = \arg \max_y P(x | y) \quad \text{ML (maximum likelihood).}$$

The two special cases: MAP and ML

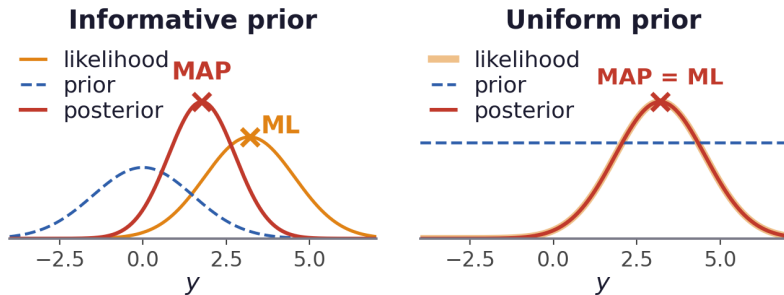


Figure 6: An informative prior pulls the posterior peak away from the likelihood peak, so MAP and ML differ. A uniform prior leaves the posterior proportional to the likelihood and the two coincide.

Continuous labels: squared loss and absolute loss

When y is a real number, such as depth or orientation, the loss is a function of the error and the Bayes rule is an estimator.

For the squared loss $L(a, y) = (a - y)^2$, differentiate $C(a | x) = \int (a - y)^2 P(y | x) dy$ and set the derivative to zero,

$$2 \int (a - y) P(y | x) dy = 0 \implies \hat{\alpha}(x) = \mathbb{E}[y | x],$$

the posterior mean. For the absolute loss $L(a, y) = |a - y|$ the derivative is $P(y < a | x) - P(y > a | x)$, which vanishes at the posterior median.

The same posterior therefore yields different answers under different losses. Squared loss is sensitive to the tails of the posterior, so a bimodal posterior can produce an estimate lying between the two modes, at a value the posterior itself considers improbable.

Targets, distractors, and four types of outcomes

By convention $y = +1$ is the target and $y = -1$ the distractor. A decision and a truth combine into four outcomes:

- ▶ true positive and true negative, the two correct outcomes;
- ▶ false positive: you say target, the truth is distractor;
- ▶ false negative: you say distractor, the truth is target.

Write T_p, T_n, F_p, F_n for the losses of the four outcomes. Lowering one of the two error rates raises the other, because both are controlled by where the decision boundary is placed.

Targets, distractors, and four types of outcomes

		Truth	
		$y = +1$	$y = -1$
Decision	$\alpha(x) = +1$	True positive $L = T_p$	False positive $L = F_p$
	$\alpha(x) = -1$	False negative $L = F_n$	True negative $L = T_n$

Figure 7: The outcome depends on the likelihood and the loss function.

Worked example: why a positive test rarely means disease

A test with 90% sensitivity and 95% specificity, for a disease 1 person in 100 has. Out of 10,000 people, 100 have the disease and 9,900 do not:

	test +	test -	total
disease	90	10	100
no disease	495	9405	9900
total	585	9415	10000

so $P(\text{disease} | +) = 90/585 = 0.154$. In symbols,

$$\frac{0.90 \times 0.01}{0.90 \times 0.01 + 0.05 \times 0.99} = \frac{0.0090}{0.0090 + 0.0495} = 0.154.$$

When the disease is rare, most positive tests come from the healthy group: the 9,900 healthy people produce 495 false positives, five times the number of true positives available.

Worked example: why a positive test rarely means disease

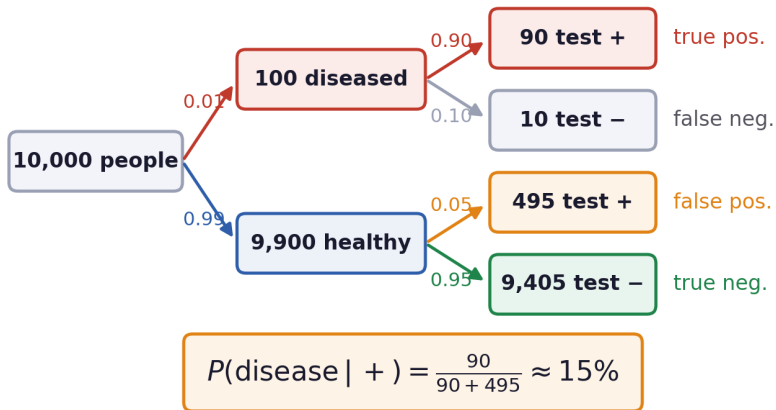


Figure 8: Where the positive tests come from.

The Bayes rule is a log-likelihood ratio test

For two classes there are two candidate answers. Write p_+ and p_- for the two posteriors $P(y = +1 | x)$ and $P(y = -1 | x)$; the Bayes rule decides +1 when $C(+1 | x) < C(-1 | x)$, which is

$$T_p p_+ + F_p p_- < F_n p_+ + T_n p_-.$$

Collecting the two posteriors and dividing gives

$$\frac{P(y = +1 | x)}{P(y = -1 | x)} > \frac{T_n - F_p}{T_p - F_n},$$

and substituting Bayes' theorem, where $P(x)$ cancels, leaves a test on the log-likelihood ratio,

$$\Lambda(x) = \log \frac{P(x | y = +1)}{P(x | y = -1)} > T.$$

The scalar T is a function of loss and prior.

The Bayes rule is a log-likelihood ratio test

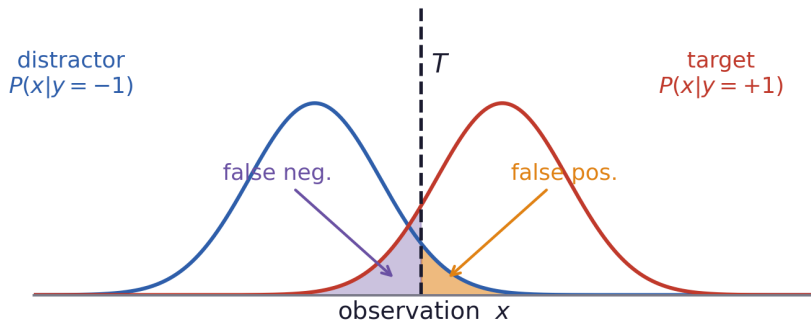


Figure 9: The threshold T on the observation axis splits it into the region where the rule answers target and the region where it answers distractor.

Where the threshold comes from

$$\underbrace{\log \frac{P(x | y = +1)}{P(x | y = -1)}}_{\text{evidence}} > \underbrace{\log \frac{T_n - F_p}{T_p - F_n}}_{\text{the loss term}} + \underbrace{\log \frac{P(y = -1)}{P(y = +1)}}_{\text{the prior term}}$$

- ▶ **Evidence.** What this observation says. Computed from the two likelihoods alone.
- ▶ **The loss term.** How much worse a miss is than a false alarm. This lowers T when misses are costly.
- ▶ **The prior term.** How rare the target was before you looked. This raises T when targets are rare.

Only the left side depends on the observation. The right side is a constant fixed before any data arrives, which is why a single scalar T suffices.

Deriving the threshold from the two costs

Start from the posterior ratio test and apply Bayes' theorem to both posteriors:

$$\frac{P(y = +1 | x)}{P(y = -1 | x)} = \frac{P(x | y = +1)}{P(x | y = -1)} \cdot \frac{P(y = +1)}{P(y = -1)} > \frac{T_n - F_p}{T_p - F_n}.$$

Move the prior ratio to the right and take logarithms, which preserves the inequality because log is increasing:

$$\log \frac{P(x | y = +1)}{P(x | y = -1)} > \log \frac{T_n - F_p}{T_p - F_n} + \log \frac{P(y = -1)}{P(y = +1)} = T.$$

With the 0–1 loss the correct outcomes cost nothing and the two errors cost one, $T_p = T_n = 0$ and $F_p = F_n = 1$, so the loss term is $\log \frac{0-1}{0-1} = \log 1 = 0$ and T reduces to the log prior ratio. With a uniform prior as well, $T = 0$ and the rule compares the two likelihoods directly.

Moving the threshold, and what it costs

Write F for the false positive rate and H for the hit rate,

$$F(T) = \int_{x^*(T)}^{\infty} P(x | y = -1) dx, \quad H(T) = \int_{x^*(T)}^{\infty} P(x | y = +1) dx,$$

where $x^*(T)$ is the observation at which $\Lambda(x) = T$. Both are decreasing in T , so no threshold lowers them together. Moving T trades one shaded area for the other.

Moving the threshold, and what it costs

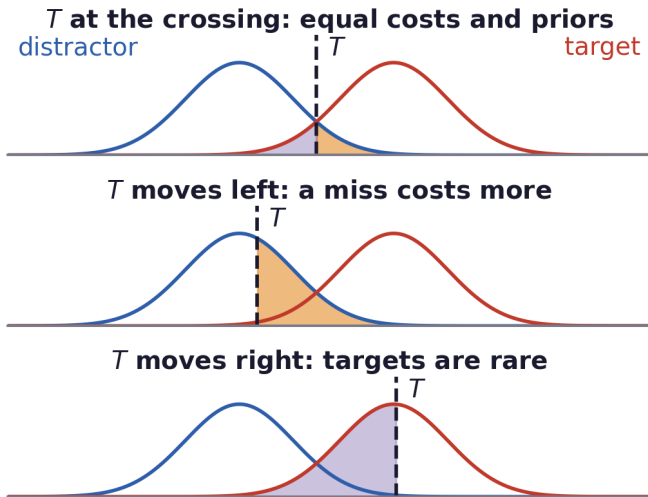


Figure 10: Raising T shrinks the orange false positive area and grows the purple false negative area. The loss and the prior decide which of the two is worth more.

Gaussian likelihoods give a quadratic decision rule

A classic book *Signal detection theory and psychophysics* (Green & Swets, 1966) modelled the observer as a Bayesian decision maker. The likelihoods are assumed Gaussian, differing in their means and in their variances,

$$P(x | y = \pm 1) = \frac{1}{\sqrt{2\pi} \sigma_{\pm 1}} \exp\left(-\frac{(x - \mu_{\pm 1})^2}{2\sigma_{\pm 1}^2}\right).$$

Substituting the two Gaussians into the log-likelihood ratio gives a quadratic polynomial in x :

$$\Lambda(x) = -\frac{(x - \mu_1)^2}{2\sigma_1^2} + \frac{(x - \mu_{-1})^2}{2\sigma_{-1}^2} + \log \frac{\sigma_{-1}}{\sigma_1}.$$

Expanding the squares collects into $\Lambda(x) = ax^2 + bx + c$ with

$$a = \frac{1}{2} \left(\frac{1}{\sigma_{-1}^2} - \frac{1}{\sigma_1^2} \right), \quad b = \frac{\mu_1}{\sigma_1^2} - \frac{\mu_{-1}}{\sigma_{-1}^2}, \quad c = \frac{\mu_{-1}^2}{2\sigma_{-1}^2} - \frac{\mu_1^2}{2\sigma_1^2} + \log \frac{\sigma_{-1}}{\sigma_1}.$$

Unequal variances give two thresholds

When $\sigma_1 \neq \sigma_{-1}$ the coefficient a is non-zero and $\Lambda(x) > T$ is a quadratic inequality. Its solution set is an interval or the complement of one, bounded by the two roots

$$x_{\pm}^* = \frac{-b \pm \sqrt{b^2 - 4a(c - T)}}{2a}.$$

With $\sigma_1 < \sigma_{-1}$ the coefficient a is negative, the parabola opens downward, and the rule answers target on the interval between the roots. An extreme observation is evidence for the class with the larger variance.

Unequal variances give two thresholds

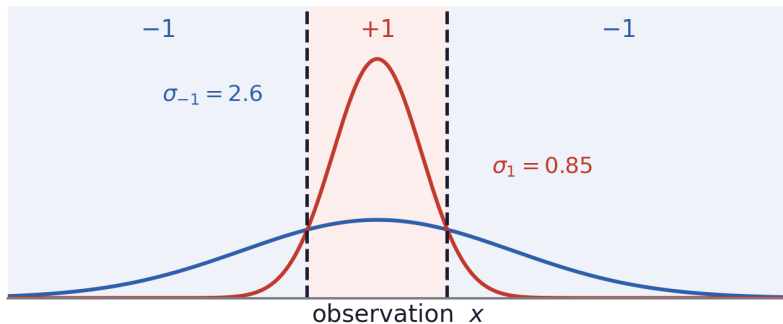


Figure 11: Two thresholds, one decision region. Far from both means the broad distractor density is larger, so the rule answers -1 on both flanks.

Equal variances: everything reduces to d'

Set $\sigma_1 = \sigma_{-1} = \sigma$. The quadratic terms cancel and the log-likelihood ratio is linear in the observation,

$$\Lambda(x) = \frac{\mu_1 - \mu_{-1}}{\sigma^2} \left(x - \frac{\mu_1 + \mu_{-1}}{2} \right).$$

Solving $\Lambda(x) > T$ for x , with $\mu_1 > \mu_{-1}$ so the coefficient is positive, gives a single threshold on the observation itself,

$$x > x^* = \frac{\mu_1 + \mu_{-1}}{2} + \frac{T\sigma^2}{\mu_1 - \mu_{-1}}.$$

Dividing the separation by σ gives the sensitivity index

$$d' = \frac{\mu_1 - \mu_{-1}}{\sigma},$$

the separation measured in units of σ .

Equal variances: everything reduces to d'

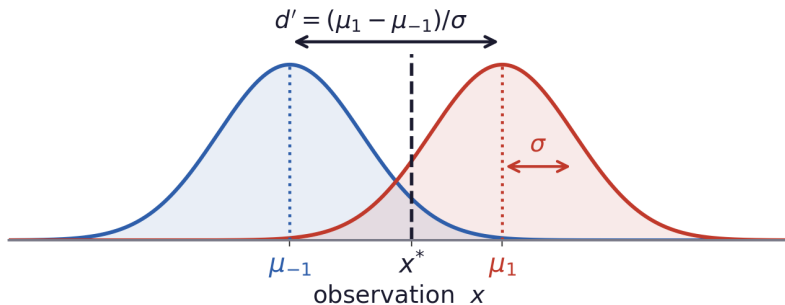


Figure 12: d' is how far apart the two densities are; T is where the threshold sits between them.

Sweeping the threshold traces the ROC curve

Fix a threshold. The fraction of the target density above it is the hit rate H , and the fraction of the distractor density above it is the false positive rate F . That pair of fractions is one point in the unit square, and sweeping T traces the receiver operating characteristic (ROC) curve.

- ▶ A low threshold keeps almost all of both densities, so H and F are both near one.
- ▶ A high threshold keeps almost none of either, so both are near zero.
- ▶ d' fixes which curve you are on, and T picks the point on it.

The two ends of the curve are the same for every pair of densities. What changes with d' is how far the curve bows away from the diagonal in between.

Sweeping the threshold traces the ROC curve

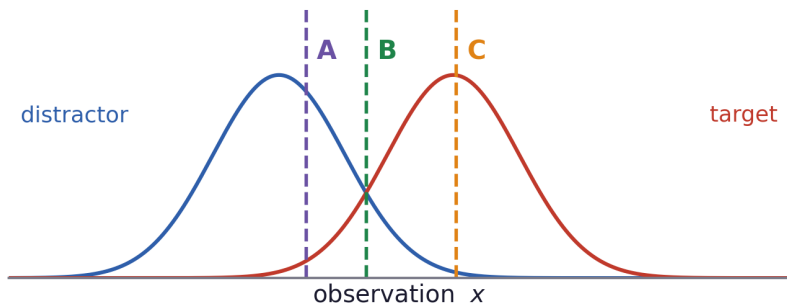


Figure 13: Three thresholds on the same pair of densities. A keeps almost all of both, C keeps almost none of either, and B sits between them.

The ROC curve in closed form for equal variances

With equal variances and Φ the standard normal cumulative distribution, the two rates at threshold x^* are

$$F = 1 - \Phi\left(\frac{x^* - \mu_{-1}}{\sigma}\right), \quad H = 1 - \Phi\left(\frac{x^* - \mu_1}{\sigma}\right).$$

Eliminating x^* between them gives the curve

$$H = 1 - \Phi\left(\Phi^{-1}(1 - F) - d'\right),$$

which depends on the two Gaussians only through d' . Every threshold on the same pair of densities lands on this one curve, and a larger d' bows it further towards the top left corner.

The ROC curve in closed form for equal variances

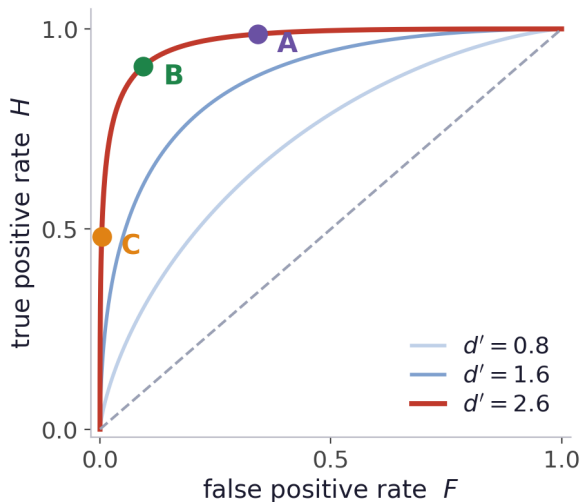


Figure 14: One curve per value of d' , one point per threshold. The diagonal is $d' = 0$, where the observation carries nothing about the class.

Reading d' off the ROC curve

Define $z(p) = \Phi^{-1}(p)$, the inverse of the standard normal cumulative distribution: $z(p)$ is the number of standard deviations below which a fraction p of a standard normal lies. Applying z to the two rates,

$$z(H) = \frac{\mu_1 - x^*}{\sigma}, \quad z(F) = \frac{\mu_{-1} - x^*}{\sigma},$$

and subtracting removes the unknown threshold,

$$z(H) - z(F) = \frac{\mu_1 - \mu_{-1}}{\sigma} = d'.$$

A single observed pair (F, H) therefore estimates d' , which is how psychophysical experiments report sensitivity independently of how cautious the observer was. Plotted on z axes the ROC curve becomes a straight line of unit slope and intercept d' , and a departure from unit slope indicates unequal variances.

The Bayes risk of an equal-variance detector

With the 0–1 loss and equal priors, $T = 0$, so x^* is the midpoint $(\mu_1 + \mu_{-1})/2$ and both error rates are equal,

$$F = 1 - \Phi\left(\frac{d'}{2}\right) = \Phi\left(-\frac{d'}{2}\right), \quad 1 - H = \Phi\left(-\frac{d'}{2}\right).$$

The Bayes risk is the average of the two,

$$R(\hat{\alpha}) = \Phi\left(-\frac{d'}{2}\right).$$

At $d' = 1$ this is 0.31, at $d' = 2$ it is 0.16, at $d' = 4$ it is 0.023. The risk falls towards zero as the two densities separate, and it reaches zero only in the limit, which is the concrete form of the statement that the Bayes risk is never zero when the classes overlap.

Where the theory came from

Wartime code-breaking and radar; then psychophysics; then machine learning. The same log-likelihood ratio test appears in all three, under different names and with different quantities playing the part of x . For radar the observation is a voltage on a screen, and the threshold is set by how much worse a missed aircraft is than a spurious alarm. For psychophysics the observation is an internal response, and d' and T are estimated from a subject's hit and false positive rates. For a classifier the observation is an image or a feature vector, and the threshold moves with the class frequencies in the training set.

Where the theory came from

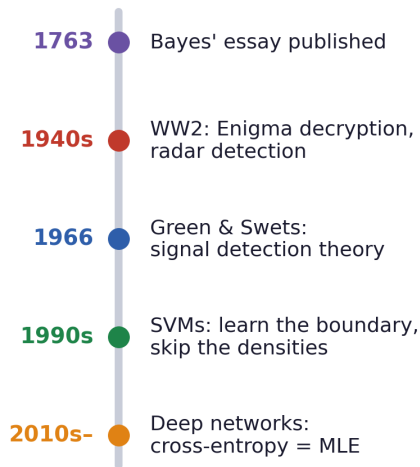


Figure 15: One test, three fields. Turing and Good used log-likelihood ratios against Enigma while radar engineers were setting detection thresholds; Green and Swets brought the analysis to human observers; the cross-entropy loss carried it into deep networks.

From hand-designed distributions to learned ones

Until now $P(x|y)$ and $P(y)$ were assumed known. Assume a functional form $P(x|\gamma)$ and estimate γ from a set D of N observed samples. Assume too that the data are independent and identically distributed (i.i.d.), so that

$$P(D|\gamma) = \prod_{i=1}^N P(x_i|\gamma).$$

$P(D|\gamma)$ is the probability of the whole data set under a particular γ . Estimating γ means picking the value that makes the observed data most probable,

$$\hat{\gamma} = \arg \max_{\gamma} \prod_{i=1}^N P(x_i|\gamma) = \arg \max_{\gamma} \sum_{i=1}^N \log P(x_i|\gamma),$$

where the logarithm turns the product into a sum and leaves the maximiser unchanged.

Maximum likelihood, and what a prior adds

Treating γ as the unknown of a decision problem, with the 0–1 loss and a prior $P(\gamma)$, the MAP estimate is

$$\hat{\gamma} = \arg \max_{\gamma} \left[\sum_{i=1}^N \log P(x_i | \gamma) + \log P(\gamma) \right].$$

ML is Bayes decision theory with a uniform prior over γ and all errors penalised equally. One prior term against N data terms, so the prior's influence shrinks like $1/N$.

You can learn by only a few examples with a sophisticated prior, which is the regime cognitive science cares about: a child sees a handful of dogs and generalises, because the prior over what kinds of categories exist is already rich.

Maximum likelihood for a Gaussian

Take $P(x | \gamma) = \mathcal{N}(x; \mu, \sigma^2)$ with $\gamma = (\mu, \sigma^2)$. The log-likelihood of N i.i.d. samples is

$$\log P(D | \gamma) = -\frac{N}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^N (x_i - \mu)^2.$$

Differentiating with respect to μ and setting the result to zero,

$$\frac{1}{\sigma^2} \sum_{i=1}^N (x_i - \mu) = 0 \implies \hat{\mu} = \frac{1}{N} \sum_{i=1}^N x_i,$$

and differentiating with respect to σ^2 ,

$$-\frac{N}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_{i=1}^N (x_i - \mu)^2 = 0 \implies \hat{\sigma}^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \hat{\mu})^2.$$

The sample mean and the sample variance are the maximum likelihood estimates. The variance estimate is biased low by a factor $(N-1)/N$, because the same data supplied $\hat{\mu}$.

A Gaussian prior adds a fixed number of observations

Keep σ^2 known and put a Gaussian prior $\mathcal{N}(\mu; \mu_0, \tau^2)$ on the mean. The log posterior is

$$-\frac{1}{2\sigma^2} \sum_{i=1}^N (x_i - \mu)^2 - \frac{(\mu - \mu_0)^2}{2\tau^2} + \text{const},$$

a quadratic in μ , so the posterior is Gaussian. Differentiating and solving gives

$$\hat{\mu}_{\text{MAP}} = \frac{\frac{N}{\sigma^2} \bar{x} + \frac{1}{\tau^2} \mu_0}{\frac{N}{\sigma^2} + \frac{1}{\tau^2}}, \quad \frac{1}{\sigma_{\text{post}}^2} = \frac{N}{\sigma^2} + \frac{1}{\tau^2}.$$

Precisions add: the data contribute N/σ^2 and the prior $1/\tau^2$. The estimate is a weighted average of the sample mean and the prior mean, and the prior carries the weight of σ^2/τ^2 observations. As N grows, the data term dominates and $\hat{\mu}_{\text{MAP}} \rightarrow \bar{x}$.

Evidence accumulates and the prior's influence falls

With one or three observations the posterior is broad, and the prior still shifts where it sits. By forty observations the data has decided, and the choice of prior no longer matters.

Statisticians resisted Bayesian methods because the prior is a choice; I. J. Good and Turing used them anyway. The objection weakens as data arrives, because the prior enters the posterior once and the likelihood enters N times.

Evidence accumulates and the prior's influence falls

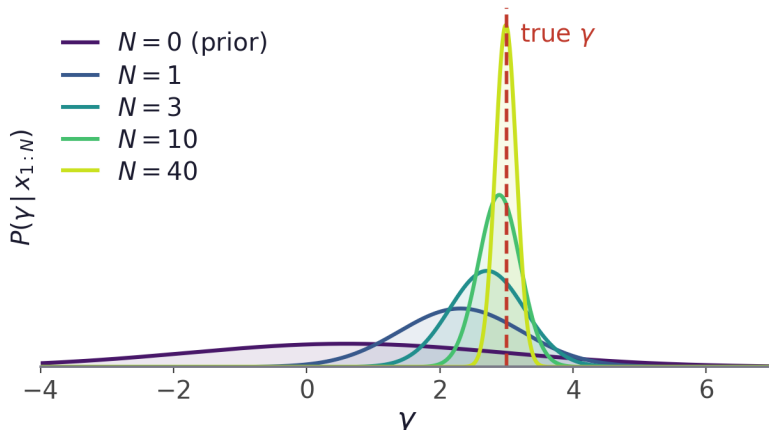


Figure 16: The posterior over γ after N observations, starting from a broad prior centred away from the truth. Each batch of data narrows it and pulls it towards the true value.

The prior matters only when data is scarce

Reading the MAP estimate of the Gaussian mean as a weighted average, the two weights are

$$w_{\text{prior}} = \frac{1/\tau^2}{N/\sigma^2 + 1/\tau^2}, \quad w_{\text{data}} = \frac{N/\sigma^2}{N/\sigma^2 + 1/\tau^2},$$

so w_{prior} falls like $1/N$. A good prior makes you learn with less data. The ratio σ^2/τ^2 measures how many observations the prior is worth. A prior as broad as the noise, $\tau = \sigma$, is worth one observation and is overwhelmed immediately; a prior ten times tighter is worth a hundred. This is why the choice of prior is contested for small data sets and irrelevant for large ones.

The prior matters only when data is scarce

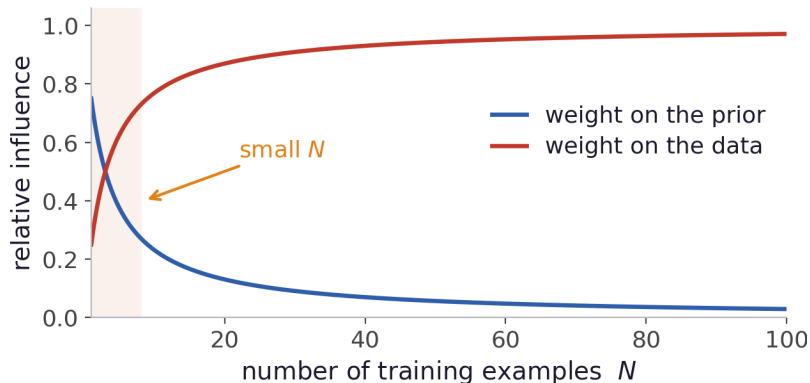


Figure 17: The two weights as a function of N . The shaded strip on the left is the regime where the prior still decides the answer, and it is the regime human learning operates in.

Learn the decision rule directly

Advocates argue that data is precious and should not be spent estimating distributions when only a rule is wanted. The true joint $P(x, y)$ is unknown, so the sample average of the loss replaces the expectation over it,

$$R_{\text{emp}}(\alpha) = \frac{1}{N} \sum_{i=1}^N L(\alpha(x_i), y_i), \quad \hat{\alpha} = \arg \min_{\alpha \in \mathcal{A}} R_{\text{emp}}(\alpha).$$

This is the risk computed under the empirical distribution $\hat{P}(x, y) = \frac{1}{N} \sum_i \delta(x - x_i) \delta(y - y_i)$, which places mass $1/N$ on each sample. The sample average converges to the true risk when the samples are i.i.d.

Learn the decision rule directly

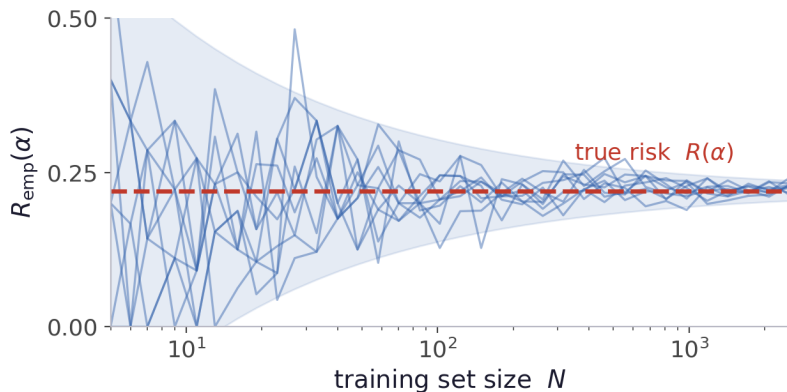


Figure 18: Nine independent training sets. The empirical risk of a fixed rule scatters around the true risk with a spread of order $1/\sqrt{N}$.

Support vector machines

The argument is that estimating probabilities is wasteful: go straight for the boundary. A support vector machine (SVM) takes linear rules $\alpha(x) = \text{sign}(w \cdot x + b)$ and, among those that separate the training data, picks the one whose margin is widest:

$$\min_{w,b} \frac{1}{2} \|w\|^2 \quad \text{subject to} \quad y_i(w \cdot x_i + b) \geq 1 \quad \text{for all } i.$$

The constraint fixes the scale, so the margin is $2/\|w\|$ and maximising it minimises $\|w\|$. The boundary is fixed by a handful of points, the support vectors: those are the points where the constraint is tight, and everything far from the margin could be deleted without changing the answer. The result is an efficient classifier that carries no model of how the data were generated.

Support vector machines

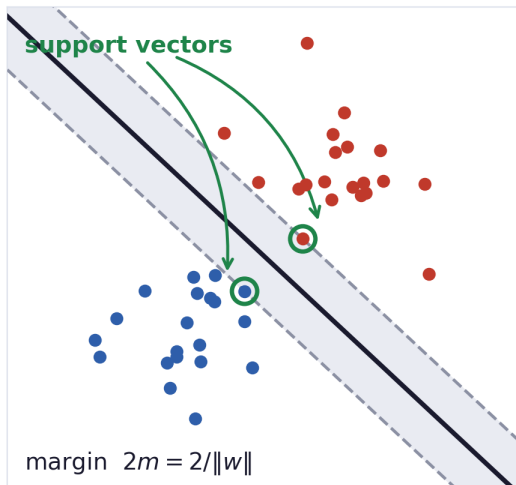


Figure 19: Spend the data on the boundary. The two dashed lines each touch one support vector, and the band between them is empty.

The soft margin and the hinge loss

Real data is rarely separable, so slack variables $\xi_i \geq 0$ allow violations at a price,

$$\min_{w, b, \xi} \frac{1}{2} \|w\|^2 + C \sum_{i=1}^N \xi_i \quad \text{subject to} \quad y_i(w \cdot x_i + b) \geq 1 - \xi_i.$$

At the optimum $\xi_i = \max(0, 1 - y_i(w \cdot x_i + b))$, so the problem is empirical risk minimisation with the hinge loss plus a penalty on $\|w\|^2$,

$$\min_{w, b} \sum_{i=1}^N \max(0, 1 - y_i(w \cdot x_i + b)) + \frac{1}{2C} \|w\|^2.$$

The hinge loss is an upper bound on the 0–1 loss that is convex, so the minimisation is tractable. The penalty term is the regularizer, and C sets how much training error is accepted in exchange for a wider margin (Cortes & Vapnik, 1995).

PAC theory: capacity and generalization

Probably approximately correct (PAC) learning bounds the gap between the empirical risk and the true risk uniformly over a rule class. For a class of Vapnik–Chervonenkis dimension h , with probability at least $1 - \eta$, every $\alpha \in \mathcal{A}$ satisfies

$$R(\alpha) \leq R_{\text{emp}}(\alpha) + \sqrt{\frac{h(\log(2N/h) + 1) - \log(\eta/4)}{N}}.$$

PAC gives an upper bound on the data needed for the estimated rule to be close to the best rule in the class. You need more data than the capacity of the rule class, and regularization reduces that capacity. The bounds are loose, and capacity is hard to measure except for simple rules (Valiant, 1984; Vapnik & Chervonenkis, 1971).

PAC theory: capacity and generalization

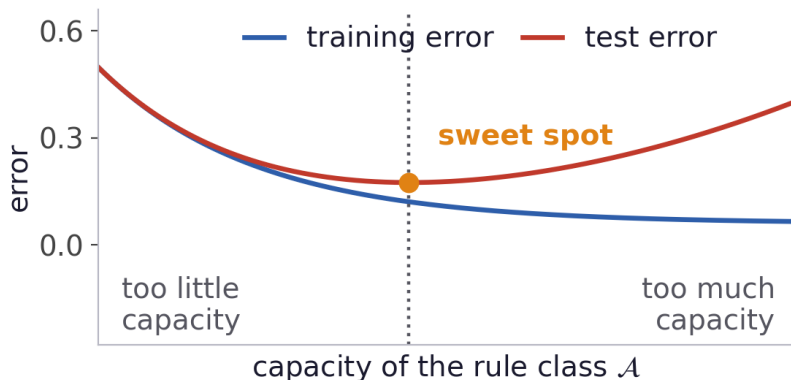


Figure 20: Training error falls with capacity while the bound on the gap grows, so the test error has a minimum.

Learning the posterior directly is what CNNs do

A convolutional neural network (CNN) is a parameterised family of distributions $P(y | x; \theta)$. With no prior on θ and equal costs, Bayes decision theory recommends maximum likelihood, which is the cross-entropy loss:

$$\hat{\theta} = \arg \max_{\theta} \sum_{i=1}^N \log P(y_i | x_i; \theta) = \arg \min_{\theta} \left[- \sum_{i=1}^N \log P(y_i | x_i; \theta) \right].$$

Asymmetric losses appear where they should: edge detection penalises a missed edge more, by weighting the two classes differently in the same sum. Over-parameterised networks can fit random labels and still generalize, which PAC capacity bounds do not explain (Zhang et al., 2017).

BDT measures performance by the average case

Every guarantee so far is a statement about $R(\alpha)$, an average over one fixed $P(x, y)$. A region of input space of probability ϵ contributes at most $\epsilon \max L$ to the risk, so a rule can fail on all of it and still be optimal. A rule can be optimal on average and still fail on every case that matters.

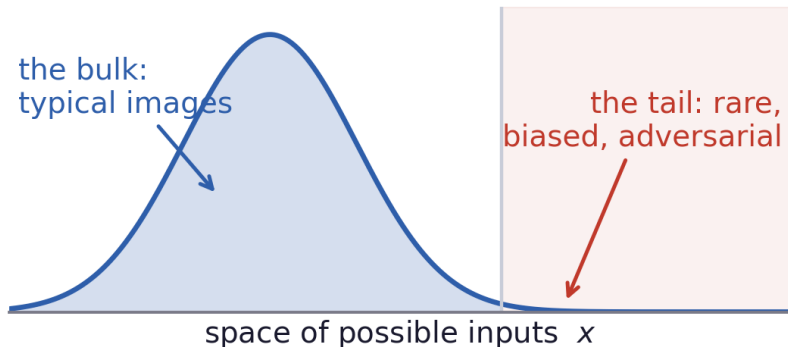


Figure 21: The risk is dominated by the bulk of the distribution. Rare inputs enter weighted by their probability, which is what makes them cheap to get wrong.

Dataset bias, and domain shift

The rule is optimal for the dataset's $P(x, y)$, and that distribution differs from the world's. Train on crowds in New York, deploy in Beijing, the algorithm would perform badly. Writing P_s for the source and P_t for the target, the rule minimises

$$R_s(\alpha) = \mathbb{E}_{P_s}[L] \quad \text{while it is judged by} \quad R_t(\alpha) = \mathbb{E}_{P_t}[L],$$

and the two are related only through how much P_s and P_t overlap. Torralba and Efros (2011) showed that a classifier can identify which dataset an image came from, which means the datasets differ from each other, and from the world, in ways a model can pick up on.

Dataset bias, and domain shift

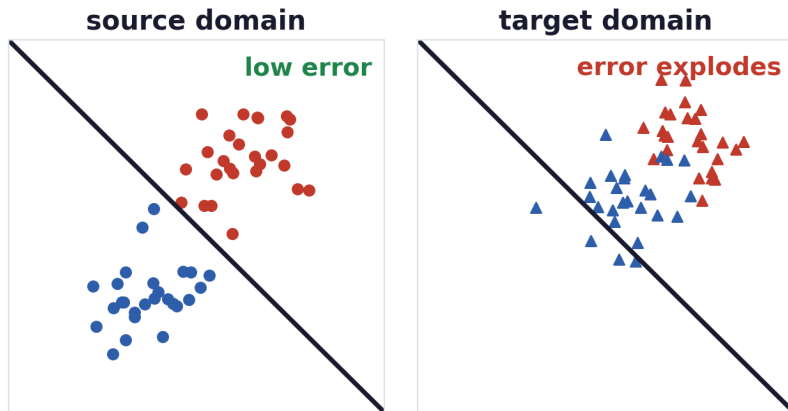


Figure 22: The boundary learned on the source applied unchanged to the target. The classes are still separable, and the inherited boundary is in the wrong place.

Searching for failure modes instead of averages

- ▶ **Average-case evaluation.** Sample a test set from the same distribution. This measures what usually happens.
- ▶ **Adversarial examination.** Search the stimulus space for inputs that make the rule fail. This measures what can happen.

Adversarial examples are perturbations too small to see that flip the decision entirely (Szegedy et al., 2014). Being vanishingly improbable, they cost almost nothing in risk: a set of probability 10^{-9} changes $R(\alpha)$ in the ninth decimal place. Average-case optimality and worst-case fragility are perfectly compatible.

Testing by search reports the worst value of the loss over a region of input space, $\max_{x \in \mathcal{X}} L(\alpha(x), y(x))$, which is a different quantity from the risk and has to be measured on purpose.

When is BDT sufficient?

If a dataset could represent every possible image, estimating $P(y | x)$ would be enough. In an open world, we will need $P(x | y)$ and $P(y)$. A generative model can be asked questions the training set never posed: what this object looks like from another viewpoint, whether an observation is unlike anything in the training distribution, how a new class relates to the old ones.

When is BDT sufficient?

Generative

learn $P(x|y)$ and $P(y)$

then $P(y|x) \propto P(x|y)P(y)$

can generate data, absorb
new classes, reason causally

costly, reusable

Discriminative

learn $P(y|x)$ or α directly

SVMs, CNNs, most deep nets

no effort spent modelling x

the test world must match

efficient, brittle

Figure 23: The two routes to a decision. The discriminative route is cheaper and assumes the test world matches the training world; the generative route pays for a model of x and can spend it elsewhere.

Summary

- ▶ Bayes decision theory has three ingredients: a joint distribution $P(x, y)$, a set of decision rules, and a loss function.
- ▶ The Bayes rule minimises the expected loss. It depends on the observation only through the posterior, which is what licenses learning the posterior directly.
- ▶ MAP and ML are the two special cases: equal costs, and equal costs with a uniform prior.
- ▶ For two classes the rule is a log-likelihood ratio test. The threshold T carries the prior and the loss, and moving it trades one kind of error for the other.
- ▶ Learning is Bayes decision theory again: maximum likelihood, the empirical risk and the cross-entropy loss all follow from it.
- ▶ The guarantees are about the average over one fixed distribution. Dataset bias, domain shift and adversarial cases live outside that average.

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